TIM: Tree Imaging Machine for high resolution rapid digital images of tree cores and cookies

October 27, 2026

Abstract

1. Studies of tree rings from climatology to ecology have become increasingly important with anthropogenic climate change, driving a need for research across more locations and species. Meeting the increasing demand requires methods to image samples rapidly and efficiently, while also producing high quality images.

2. Here we present the Tree Imaging Machine (TIM)—a do-it-yourself fully open scanning tool to digitize tree cookies and cores. TIM takes partially overlapping microscopic images of samples and stitches them together to form a mosaic, which can be zoomed in to visualize features on the scale of 0.01 mm. We provide all instructions for the open-source hardware, using readily available parts, alongside the related software.

3. With scan resolution of at least 21,000 DPI (1.2 micron per pixel), TIM produces one of the highest resolution images among similar tools for less than $3,500 CAD. Operators can prepare multiple samples and let the machine work until the queue is finished. Given this batch approach and automatic stitching, TIM provides a major speed advance for high-resolution image capture of tree rings, and can rapidly digitize cores of over 50 cm and complete cookies.

4. TIM provides a new method to gather high quality tree ring images at larger sizes and dimensions, integrating higher scanning resolution, tree cookie sampling, and batch digitization in a single low-cost package. Designed as completely open-

source, we hope TIM will facilitate the widespread sampling of tree rings across diverse ecosystems and reduce barriers for researchers with smaller budgets.

1 Introduction

Incremental growth rings from trees have provided valuable insights into the abiotic and biotic environment across multiple fields—from climatology to forestry and ecology. Dated tree rings from dendrochronology have been critical to reconstructing past climates of regional and local environments, providing the most accurate, precise, and reliable dating among alternatives (Mann *et al.*, 1999), and thus informing our understanding of climate change today (Fritts, 1971; Williams *et al.*, 2010; Guibal & Guiot, 2021; Sheppard, 2010; Williams *et al.*, 2020). Tree rings are also increasingly used in ecology to understand how plant competition affects tree growth in addition to climate (Buechling *et al.*, 2017) and have implications in forestry to managing stand dynamics (Canham *et al.*, 2004). These fields have leveraged two basic types of tree sampling techniques to capture the variation of tree rings: cores, which are cylindrical samples collected using an incremental borer, and 'cookies,' which are entire cross sections (and thus allow measuring as many radii from the sample as desired, Speer, 2010). As understanding tree growth and managing forests becomes more critical to mitigating climate change, scaling up the collection and processing of tree ring samples has become increasingly important.

The first computer-based method to measure tree ring widths used a microcomputer, a stage micrometer, and a push button, and is still used today. To acquire data, a trained technician shifts a core underneath a microscope objective and pushes a button connected to a microcomputer to record a ring's location (Robinson & Evans, 1980). This method has high precision, but depends heavily on specific technicians and maintains no image record, preventing later measurement reproducibility and uncertainty quantification (Levanič, 2007).

Image analysis of tree rings was later implemented to reduce both errors in sampling and sampling bias across individual technicians, with major methods developed in the 1980s and 1990s to digitize tree ring samples using optical scanning (McMillin, 1982;

Guay *et al.*, 1992). Optical scanners are still readily available for purchase; using a top of the line scanner (e.g., Epson Perfection V850 Pro) produces a maximum resolutions of 6,400 dpi and scan area of up to 21.6 cm x 29.7 cm. More recent techniques have aimed to achieve higher resolutions and/or larger samples through a different digitization approach—using an image-capturing system. These systems rely on a digital camera to take images, motors to move the sample or camera, a computer to manage the capturing, and a second, more powerful, computer to stitch images into a larger composite image. Decreasing the field of view of each captured image increases the resolution of the final stitched image without restricting the size of the sample (Muhlich *et al.*, 2022). Stitching multiple images into one mosaic is a common technique used in other fields such as mineralogy and cellular biology (Ro & Kim, 2021), which was first implemented for tree rings through the Advanced Tree-Ring Image Capturing System (ATRICS) (Levanič, 2007).

Since ATRICS was introduced in 2007, image capture and stitching of tree rings have become more common. Today CaptuRING offers a more modern do-it-yourself alternative to ATRICS, creating scans up to 5,339 dpi using a DSLR camera and motorized stage (García-Hidalgo *et al.*, 2022). Another alternative, Gigapixel, achieves an impressive 19,812 dpi with the same DSLR image capture and composite stitch method (Griffin *et al.*, 2021). While CaptuRING and Gigapixel both use DSLR cameras and motorized movement, they differ on two fronts. First, Gigapixel's camera is traversed across a stationary sample while CaptuRING moves a core sample underneath a stationary camera. With a mobile camera, multiple samples can be recorded sequentially without needing to remove a sample from a stage upon image completion—significantly reducing the active setup time for an operator. Second, Gigapixel automatically stitches individual images together once imaging is complete. This removes the need to transfer images to a more powerful computer for stitching as is necessary with CaptuRING. Additionally, both of these systems are designed to capture images along a singular axis as the entire width of a core fits within a single image's field of view. This assumption fundamentally restricts the digitization only to cores, as cookies have a two dimensional scanning surface.

To address some of the current limitations, while building upon the open-source and open-hardware philosophies of CaptuRING, we designed the Tree Imaging Machine (TIM). We added support to digitize both cookies and cores, increased the maximum sample length, perform image stitching directly without the need for manual file transfer, and capture batches of samples sequentially—all while minimizing cost. Our design requires the use of 3D printed components and common hand tools such as hex keys and wrenches. No specialized equipment or power tools are required beyond this. Excluding 3D printed parts, the total cost of the machine is approximately $3,500 CAD (compared to the $70,000 USD of the Gigapixel, Griffin *et al.*, 2021). We include a table detailing the cost of each component in the supplemental (Table S1). We chose to use a Raspberry Pi HQ camera with a microscope lens to save on purchasing a professional camera with a macro lens. Professional camera setups are comparable in price to the entire cost of TIM. These savings combined with not needing to purchase a professional stitching software license (such as PTGui from New House Internet Services BV, Rotterdam, NL, which is the suggested software to use with CaptuRING) makes TIM a more economical option than most products currently available. Additionally, the design is open-source and open-hardware allowing for contributors to add new sensors or alter the control methods.

2 Methods

2.1 TIM System Overview

TIM can be thought of as a combination of multiple subsystems: a gantry machine, camera, and computer (Figure 1). For the gantry machine, cartesian movement in the X, Y and Z directions is a result of two machine kits and a motor controller from OpenBuilds—the ACRO 1010, the NEMA 17 lead screw linear actuator, and the X32 Motor Controller running Grbl firmware. The machine kit includes extruded aluminum rails, carriages to slide along the rails, stepper motors to control the X and Y movement, and all the hardware to assemble the machine. To fit on the end-effector plates of the ACRO system, we made an adapter to connect the linear actuator to the X and Y axis,

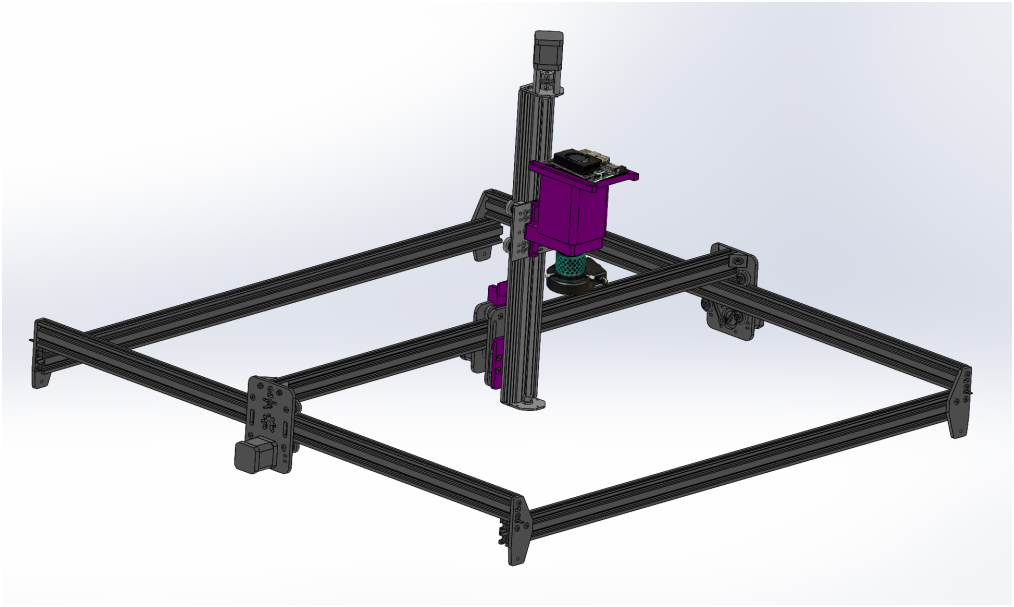


Figure 1: Complete assembly of TIM. The ACRO kit is shown in gray. Camera and computer adapters are shown in purple. All of the 3D printed parts are available for download on the NIH 3D printing repository (HYPERLINK REMOVED) and the software is open source, with instructions to recreate this tool available on GitHub (HYPERLINK REMOVED). Links to images produced by TIM are provided in Table S4.

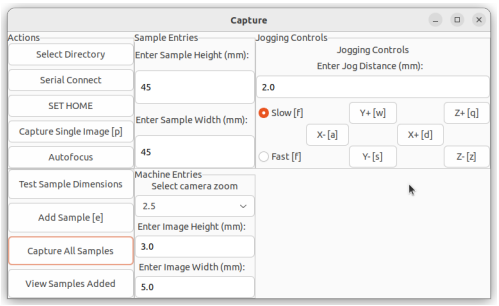
thus creating motion in the Z axis. The size of the work plane is ultimately up to the size of the ACRO kit that is purchased—allowing for more flexibility. Building on top of kits allowed for quick assembly with sound instructions and saved development time. For the camera, we chose a 12MP Raspberry Pi HQ Camera equipped with a SEEED studio microscope lens with adjustable focal length resulting in up to 150x zoom. The focal length was adjusted to make the field of view approximately 3 mm by 5 mm height and width. The camera lens has three attachment points for a microscope ring light LED. We attached a generic, white light LED ring light similar to the AmScope LED-78. The camera was connected to the gantry machine using custom, 3D printed, adapters. For the computer, we implemented all of the software control, image stitching, and graphical user interface (GUI) to run on an NVIDIA Jetson Orin Nano 8GB (Jetson edge computer), which also connects to the camera via a MIPI cable.

By choosing this combination, we were able to reduce the weight and cost of the camera significantly. This minimized the design challenges of stabilizing the movement of a heavier DSLR and allowed us to invest more in a powerful computer that can handle

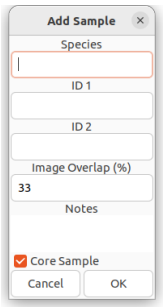
intensive image processing. The Jetson is an edge computer which drives a monitor for a GUI, sends commands to the motor controller to move the machine, runs image processing calculations for machine control, and performs calculations to stitch individual images into one mosaic. Screenshots of the GUI windows are shown in Figure 2.

2.2 Preparing Cookies and Cores

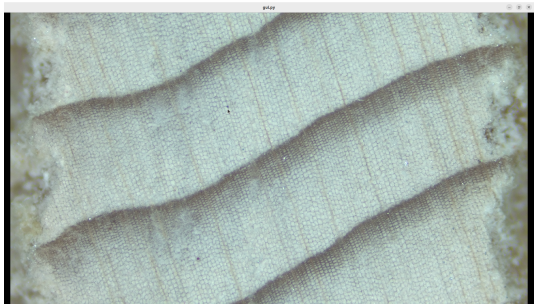
TIM can scan both tree cookies and cores but they need to be well prepared before scanning. Samples that are not well sanded may not have enough detail when stitching images together into the complete mosaic. Thus, users of TIM should follow general guidelines for preparing cookies and cores—sanding samples with incrementally increasing sand paper grit. Samples must be uniformly sanded and have a planar surface for scanning (Speer, 2010). We sanded samples using a 4 inch belt sander with lower grits between 120 and 240 for the bulk of the material removal, followed by sanding using an orbital sander with progressively finer grits between 240 and 400. We then used a microscope to visually inspect the quality of the sanding. Samples were considered to be sufficiently prepared when vessels were easily recognizable.



(a) GUI



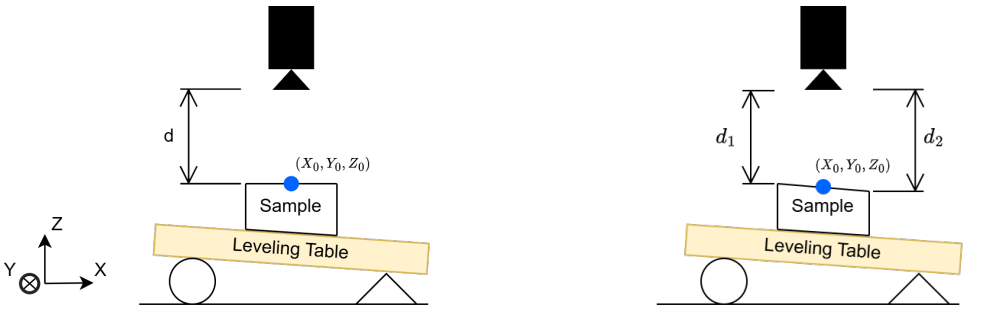
(b) 'Add Sample' dialog



(c) Live camera feed window

Figure 2: TIM is controlled using a GUI, popup windows, and a live camera feed.

TIM requires additional sanding considerations for each sample type. For cookies, it is important to have the top and bottom surfaces nearly parallel. Small differences in plane angle can be corrected using the 3D printed leveling table we designed (Figure 3a). For cores, care should be taken to maximize the scanning surface. This means removing material to be coincident to the center of the cross section of the core. Similarly to cookies, the scanning surface should be as close to parallel to the XY-plane as possible, but generally does not need to be leveled using the leveling tables.



(a) Ideal sample leveling

(b) Realistic sample leveling

Figure 3: Side view of the camera (top, in black) and sample on top of a leveling table. The ideal sample leveling shows a uniform distance, d, at all XY coordinates on the sample. This is impossible to achieve in reality, the true sample leveling has a non-uniform distance at unique XY.

2.3 Scanning Samples

TIM can scan cores, cookies, and sections of cookies. Each one of these scans requires a slightly different process, which we describe in separate subsections. A few principles are common regardless of what is being scanned, and we describe these first.

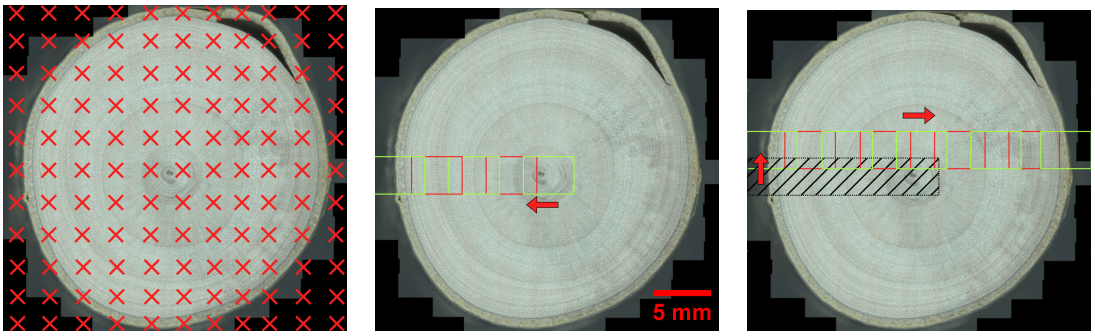
2.3.1 Regions of Overlap for Stitching

The individual images from TIM have a field of view of approximately 3 mm x 5 mm. Because cores and cookies can be many orders of magnitude larger than one of these images, the final scan of these samples will be the combination of many images stitched together. Stitching two images together requires both images to share a region of overlap in which the subject of the image is the same (Mohammadi *et al.*, 2024). Unique features inside this region of overlap act as a map to connect the two images. By default TIM is

set to have a 33% overlap with its neighbor but this can be adjusted easily while adding a sample in the GUI. The imaging process of a cookie is shown in Figure 4.

2.3.2 Golden Section Search Autofocusing

Most microscope lenses are a fixed focal length, requiring the subject to be moved in relation to the lens for focusing. Additionally, inexpensive microscope lenses often have a very small depth of field. Variations of height across the length of a sanded sample result in images losing focus during a scan (Figure 3b). An autofocusing algorithm was implemented using a Golden Section Search method as described by Gu *et al.* (2015) to capture in-focus images. In short, the camera's position is moved in the Z axis and stops when it maximizes the sharpness metric of the image. We found the variance of Laplacian sharpness metric to be easy to implement in Python and robust across samples (Pech-Pacheco *et al.*, 2000).



(a) (b) (c)

Figure 4: An example imaging process of a cookie sample in two dimensions. Here, the height and width of the capture region enclose the entire surface area of the cookie. TIM begins by generating a grid of target coordinates to systematically capture a grid of continuous and partially overlapping images (a). TIM takes its first image at the center coordinate of the sample, then the rectangular field of view from the camera moves to adjacent target coordinates until it reaches the left boundary—capturing an image for each target coordinate (b). After completing the row, TIM moves to a new row and continues capturing adjacent images (c).

2.3.3 Core Scanning Procedure

Scanning a core starts with aligning the length of the sample to the Y-axis of the machine at any X position. We 3D printed a tray to hold multiple cores next to each other to make

the alignment process faster. From there, the operator needs to navigate the camera to an in-focus position anywhere along the core and enter basic metadata such as sample ID. Multiple cores can be put next to each other and added to a queue to be sequentially scanned for operational convenience.

A button on the GUI starts the procedure to scan all of the samples that were added to the queue. The machine first moves to the position that was recorded when the operator added the sample. It then moves up the length of the core in the positive direction of the Y-axis, capturing overlapping images until the top edge of the core is captured. TIM then returns back to the starting location, repeating the imaging procedure in the negative direction of the Y-axis until encountering the bottom edge of the core.

The top and bottom edges of the core are detected by comparing the sharpness metric to a configurable threshold. When the camera moves past either edge of a core, there is no subject in the focused plane of the lens. These images are very blurry and the sharpness metric is at least an order of magnitude less than the images that are focused on the core. If the Golden Section Search algorithm cannot find an image with a higher sharpness metric than this threshold, it is safe to assume the sample is no longer in the frame. This technique reduces scanning setup time as the operator does not need to measure the length of each core when adding the sample into scanning queue. Among the 20 species (gymnosperms and angiosperms) that we have scanned, we found the sharpness metric did not vary significantly when in-focus, allowing multiple similar species to be scanned in the same batch.

Golden Section Search autofocusing is robust, but takes up to 10 seconds to obtain the most in-focus image (in our implementation). When capturing tens or hundreds of images per sample, autofocusing time thus becomes significant. In the interest of minimizing scanning time per sample, we chose to only autofocus when necessary. Autofocusing occurs on the very first image to get a baseline sharpness metric. The machine then moves to capture the neighboring image. If the new image has at least a sharpness metric of 95% of its previously imaged neighbor, we choose not to run the autofocusing procedure. Otherwise the system autofocuses. If the prepared surface of the core is

perfectly planar to the table that TIM is installed on, we have found that it is possible to only autofocus once for tens of images while maintaining high image sharpness (see Table S4). This approach of auto-focusing only when deviations from the previous sharpness are found ensures that all neighboring images have functionally the same sharpness, while minimizing the time spent on autofocusing. While scanning well prepared samples, we generally saw an autofocusing routine for every 5-10 images. All of the images captured with this method maintained high image sharpness throughout the sample and were well suited for stitching. This time-saving feature can be tuned in the configuration file to have a more stringent or lenient sharpness threshold. The autofocusing procedure will run more frequently when the sample is not aligned well to the camera and in cases where the sample is less even. However, samples that have visible scoring marks from sanding, noticeable non-planar surfaces, or other major surface irregularities will likely have difficulties scanning with or without adaptive autofocusing. We have found that such samples simply require more attentive sanding to ensure a blemish free, planar surface for successful and rapid scanning.

2.3.4 Accounting for Translation Error Between Core Samples

Movement accuracy issues arise when more than one core is added to the digitization queue. While the ACRO has a theoretical 4.5 micron translational accuracy, the machine still sometimes over- or under-shoots the true location when moving to a stored location of the core—resulting in the core not being centered in the frame. We thus implemented a centering procedure that realigns the center of the core to the center of the image frame when the machine moves to a new sample.

The goal with the recentering procedure is to prevent the scan from including significant portions of blurry background. This process works by creating a region of interest in a horizontal strip through the center of the image. The strip is further segmented into multiple subsections aligned in a row. In each of the subsections, the sharpness metric is calculated. Subsections with very low sharpness are likely to be background. If background is detected in the left most or right most subsections, corrective movements in

the X direction are made to recenter the core.

2.3.5 Cookie Scanning Procedures

Cookies can be both partially and completely scanned. Scanning entire cookies is possible and requires the operator to define a minimum bounding box using the Sample Height and Sample Width text boxes in the GUI. Additionally, the operator must navigate the camera to the center of the cookie before adding the sample to the queue. A button called "Test Cookie Boundaries" was made to trace the bounding box of the cookie for the operator to verify that the sample is truly within the bounds. These height and width bounds are used to calculate where the machine needs to move to achieve a set of partially overlapping images in both directions.

The file sizes of entire cookie scans can quickly become too large to analyze with commonly available software. A full resolution scan (21,000 DPI) of a 10 cm x 10 cm cookie is multiple gigabytes in a lossless file format. TIM can adjust the resolution of the final scan to be any percentage of the maximum DPI if needed. Whole cookie scans are interesting for archiving (though they will likely need special considerations for most samples at high resolution).

Portions of cookies can be scanned by treating them as if they were a core. This can be done by aligning the pith of the cookie's major or minor axis to the Y axis and using the core scanning procedure. Any radius of the cookie can be scanned this way.

2.3.6 Image Stitching

Image stitching is a well explored field, ranging from panoramic images taken on smart phones to highly tuned microscopy slide stitching. Stitching one image to another requires both to share a region of overlap. When capturing a sample using TIM, the operator has the ability to choose a percentage amount of overlap between images. Most samples are digitized with a percentage overlap between 33% and 50%.

After testing numerous image stitching softwares, we found the Python package Stitch2D most capable of stitching our images successfully (Mansur, 2022). This package

wraps OpenCV functions for finding distinctive image features using the scale-invariant feature transform (SIFT) method and matches these features within the overlapping region of adjacent images (Lowe, 2004). The default implementation of the package produces high quality stitches but is memory inefficient. We implemented a few key memory-conscious changes such as loading only one image into RAM at a time and writing the final image to a memory-mapped NumPy file before converting to a TIFF file. These changes allowed for the package to run without being limited by RAM on a Jetson edge computer without a problem.

Some tree ring analyses may not need the maximum scanning resolution produced by TIM. At maximum resolution, TIM's scans quickly produce large file sizes and can be costly to store at scale. To address this, a parameter in the machine configuration file allows the operator to choose a downsizing ratio to apply to images before they are stitched, decreasing the file size of the scans. With the same images, multiple final stitched resolutions can be made.

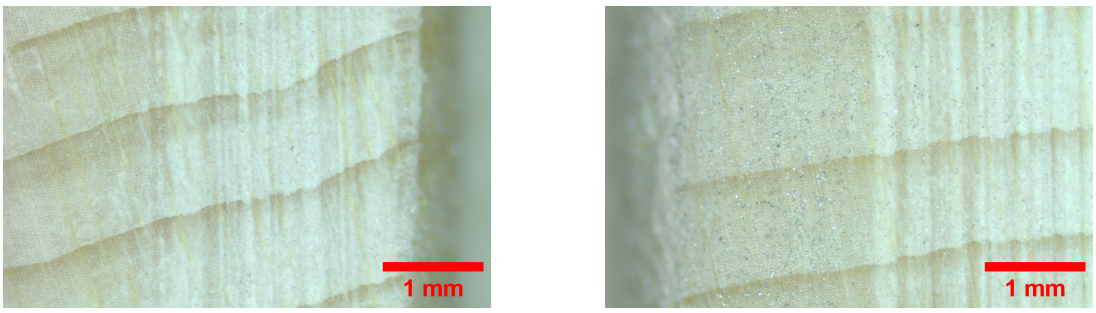
2.4 Scanning Hard-to-Stitch Samples

We found a small number of samples were difficult to stitch and developed a method to aid stitching in these cases. These cases were rare and involved either cores from (1) species with very hard to distinguish and light-colored anatomy (e.g., *Callitropsis nootkatensis*) that were not fixed with basic staining approaches, or (2) they retained scratches that we could not remove even with 1,500 grit sanding. For such cases, we developed a particle smearing method (described below) that aided the stitching process. Particle smearing does not need to be done to every sample and is best reserved for samples that have proven difficult to stitch after first trying other approaches to adjust samples (e.g., staining or additional sanding). When needed though, we found this approach very successful and that it did not interfere with the measurements of annual growth rings. Measurement of other anatomical measurements such as vessel counting, however, may be affected depending on the size of particles relative to vessels.

2.4.1 Particle Smearing

Particle smearing can be an effective technique for adding high contrast features to the prepared surface of a sample by embedding small particles into the surface of the sample. Our powder of choice was 600 mesh silicon carbide powder as it is inexpensive, available at pottery supply stores, has reliable grain sizes smaller than the finest sand paper we used, and provides good color contrast with most wood. Additionally, silicon carbide is very hard so it will not be broken into smaller grains when smeared into the surface of the sample.

No extra tools are needed for this method—a fingertip dipped in the powder is sufficient. The powder should be smeared across the sanded surface of the sample, smoothing any clumps of powder in the process. When applied correctly, trace amounts of silicon carbide will be embedded in the surface roughness of the sample which cannot be seen without microscope magnification. Most of the silicon carbide can be removed after scanning is complete by blowing compressed air on the sample. Images of a core before and after powder smearing can be seen in Figure 5.



(a) Without powder smearing (b) With powder smearing

Figure 5: A hard-to-scan *Callitropsis nootkatensis* sample due to significant scratches, with and without silicon carbide smeared on the surface of the core.

3 Results & Discussion

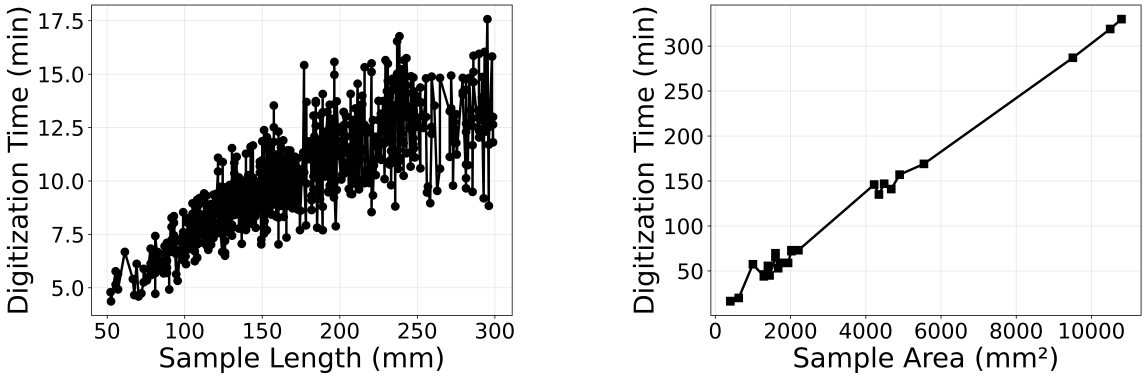
TIM is capable of producing extremely high resolution scans across diverse species. To date, we have used TIM to scan over 1,300 cores and 80 cookies spanning 20 species (Table S2). We calculated TIM can scan at least 21,000 dots per inch (DPI) by using a

photo of a 1 mm stage micrometer with 0.01 mm graduations (Figure S1) and a custom Python script. Adjusting the analog zoom ring of the camera lens changes the DPI of the scan. While there are tick marks to indicate a few zoom landmarks, the lens was not designed to make micro adjustments. Fine tuning DPI to a specific setpoint is very difficult as a result. We found the ideal zoom setting to be at the 2.5x tick mark of the available 1-4.5x range. There is still potential to zoom the lens further, at the trade off of scanning time and file size. In our analyses, we generally use 30% downscaled versions of the full resolution scans so we did not prefer the tradeoffs. Instead of committing to an exact DPI for the entire lifetime of the system, we suggest taking a photo of the slide micrometer before each batch to provide a recent DPI measurement and to update the configuration file's DPI maximum DPI field. This DPI measurement is then cast into the metadata of all the scans performed in the same batch.. When downscaling the final stitch, DPI scales linearly with the downscale percentage defined in the machine parameters. We further validated the DPI measurements by marking two control points on samples with ultra-fine tipped felt pens. The distance between the points in pixels and physical distance were measured and these control point DPI measurements were comparable to the DPI measurement from the stage micrometer (Table S5).

Digitizing samples takes significantly longer for cookies than it does for cores (Figure 6). This is due to the surface area of cookies requiring more images, as it has a squared relationship to sample radius. For large cookies, the operator can choose to selectively scan a radius of the cookie instead of the whole surface (as a single radius contains a portion of each ring) and/or scan at a lower DPI. Scanning complete cookies at maximum DPI is unrealistic as file size limits are quickly encountered with small diameter cookies.

To fit within the maximum 4 GB limits of a TIFF file, the maximum diameter of a cookie scanned at 30% of maximum resolution is limited to approximately 130 mm. This maximum cookie diameter is significantly smaller than the maximum core sample length. TIM can scan cores up to 1.5 m in length, which is 50% larger than the longest increment borer available on the market (though a larger ACRO frame than shown here would be needed to support such longer sample lengths).

Samples larger than this limit can still be scanned and stitched but TIM saves the final mosaic as a BigTIFF file. Large BigTIFF files can be opened in Python with packages such as tifffile.



(a) Core Scanning Time

(b) Cookie Scanning Time

Figure 6: Time to digitize a sample varies by sample type and scanning parameters such as percentage image overlap. Cores scan significantly faster than cookies as they do not need to take images beyond the samples edge. The samples included here were scanned between 33% and 50% overlap. The theoretical maximum sample dimensions are limited by the working plane of the machine (approximately 500 mm x 500 mm in our configuration). Sampling times are drastically influenced by configurable machine parameters such as percentage image overlap.

3.1 Strengths and Opportunities

TIM's design minimizes the barriers to build an affordable, efficient and high-resolution image capture system. We have used TIM across different operators while upgrading software incrementally over the course of two years. We found TIM works well with intermittent or intensive use. For example, we have used TIM to scan batches of 24 cores multiple times a day over the course of three months, with many batches scanned overnight. We have found that TIM does not compromise scan quality and can scan both cookies and cores. We have measured tree ring widths using the digitized samples from TIM using CooRecorder as well as ImageJ (Schneider *et al.*, 2012).

The utility of digitized samples produced by TIM could be extended through new open-data approaches. While most tree ring repositories (e.g.,the International Tree-Ring Data Bank or TreeSource's Tree-Rings Database for Canadian sites Grissino-Mayer & Fritts, 1997; Girardin *et al.*, 2021) do not store the source images of samples currently,

they provide rich metadata, which could be used to link to images stored on other more flexible repositories. Eventually extending current tree ring databases to include source images of cores and cookies (and annotating them in bulk), however, would be beneficial to the field. Finding a consistent way to archive images has the added benefit of possibly providing training data for deep learning models for vessel counting and tree ring boundary identification. Examples of such models exist and have the potential to significantly decrease the time invested in manually identifying tree anatomy (Resente *et al.*, 2021; Poláček *et al.*, 2023), and TIM could be extended to run these models while scanning (given more robust identification models).

In its current state, TIM only provides scanned images and requires users to import them into other tools for additional visual analyses, such as measuring tree-ring widths or counting vessels. A useful addition to TIM would be to automatically detect tree-ring widths or other metrics while capturing images using computer vision models. We expect this may be possible as these methods improve in computational efficiency and accuracy. Currently, we found that tree-ring detection by a recent machine learning algorithm (R-CNN presented by Poláček *et al.*, 2023) for a scanned core (23,294 x 897 pixels) took 15 minutes on a high-performance cluster and 3.5 hours on a smaller server (RTX A6000 GPU server). Although the Jetson edge computer is not as powerful as these servers, running these models in parallel to the image capturing algorithms would further reduce manual data annotation of tree-rings, and thus could help scale-up data collection.

TIM relies on purchasing a CNC machine kit from OpenBuilds which is subject to availability. Our software, however, is compatible with any cartesian CNC machine controlled by the Grbl software, a common and open-source CNC motion controller software. Many alternative CNC kits are available but would require new adapters to be made to hold the lens, camera, and computer to the machine.

4 Conclusion

By taking advantage of a common three axis cartesian machine design, powerful edge computing, and a microscope camera we were able to design a cost effective and high resolution digitization tool for wood samples. Our design significantly increases the maximum sample length and maximum resolution compared to common alternatives in the field. In addition, TIM is intended to be readily replicated by labs without significant engineering capacity and equipment. We hope that such an open-source device will facilitate the widespread sampling of tree rings across diverse ecosystems and reduce barriers for researchers with smaller budgets.

5 LLM Statement

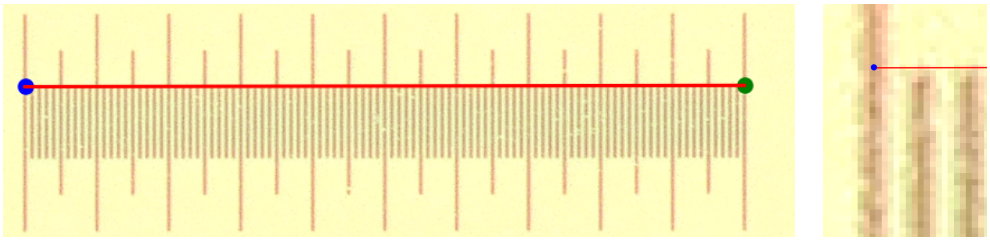
We wrote all the initial code for our first fully functioning prototype version of TIM, but used ChatGPT-4.0 to aid in implementing several improvements. Code was generated to convert our first GUI implementation to an alternative package when we faced compatibility constraints. Additionally, ChatGPT-4.0 helped quickly iterate on the tools used to manually calculate DPI using a microscope slide ruler and annotate images with guiding lines when defining ring boundaries. Other small snippets of code were generated to select pixels in specific regions of images. We carefully reviewed all generated code for functionality, and note each use of it clearly with comments in the source files.

6 Supplemental Information

6.1 How to Build TIM

TIM is intended to be operated and built by one person without having extensive machine building experience. To do this affordably, the camera and computer must attach to an off-the-shelf 3-axis CNC motion kit. Camera and computer adapters (1) need to be designed uniquely for each motion kit (which are subject to market availability). The overall price of TIM does not include the 3D printer we used to manufacture these adapters. Ordering parts to be printed can be done through online or university services. A complete list of components we used for our TIM implementation is detailed in Table S1. A simple schematic of the electronic components included in TIM is provided in Figure S2. Images detailing how to connect the camera and computer adapters to the ACRO can be found on the project's GitHub.

We are open to answering inquiries to troubleshoot specific questions of the assembly process on TIM's GitHub Issues page. At the time of publication, an external lab group has successfully replicated the TIM implementation on an ACRO.



(a) Scale Bar for DPI measurements (b) Zoomed in Scale Bar

Figure S1: Example of a single DPI measurement using a 1 mm stage micrometer with 0.01 mm graduations. We found the DPI measured at multiple locations in the field of view vertically and horizontally converged on the same value. The centerpoint of graduation marks used when taking this measurement. DPI measurements are constant so long as the lens does not change zoom. Our scans are primarily imaged with the lens' zoom dial at 2.5x out of 4.5x, resulting in approximately 21,000 DPI. There are approximately four pixels between the edges of each graduation mark, indicating that features less than 10 microns are able to be distinguished.

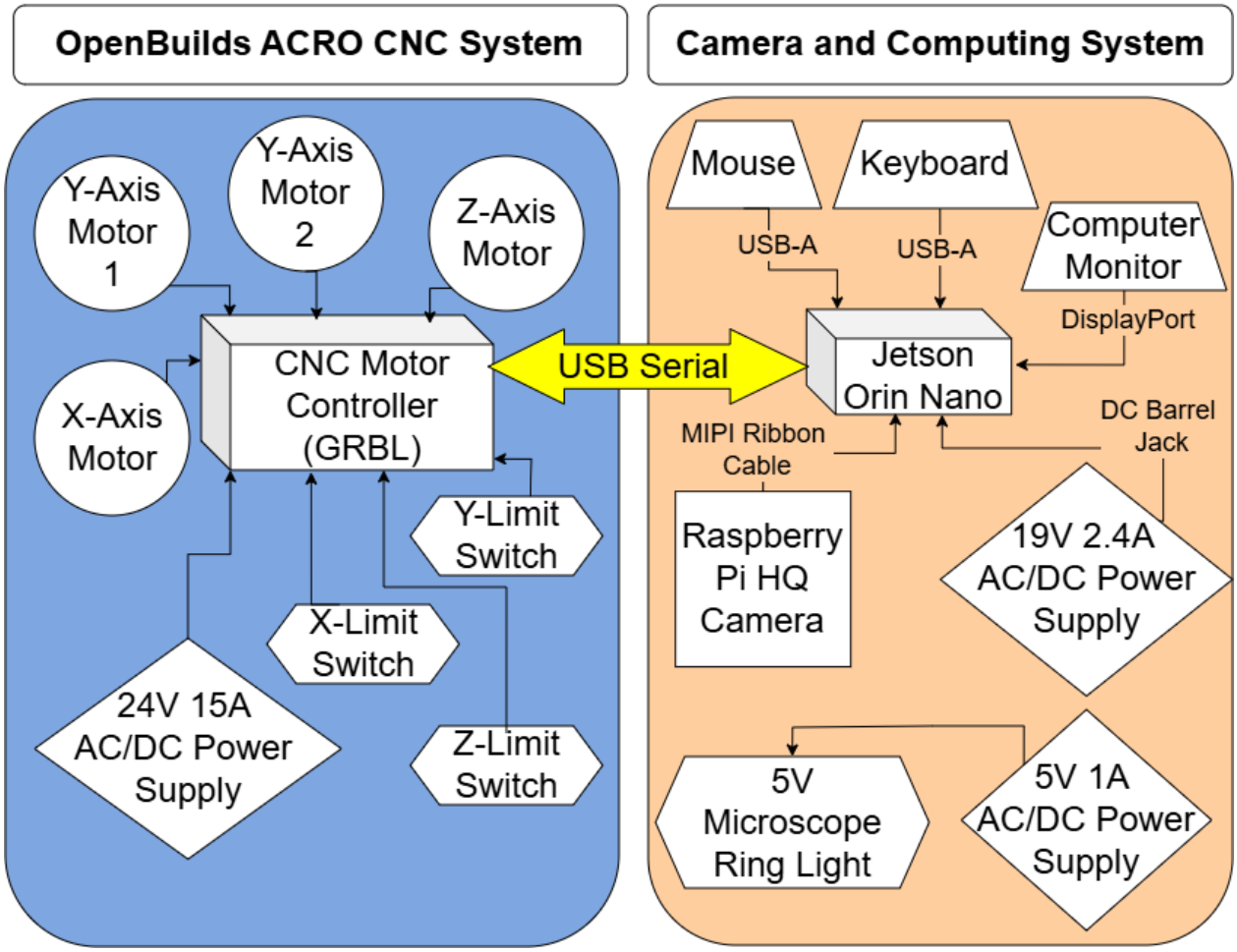


Figure S2: TIM is composed of electronic components common to many 3-axis CNC machines, with the addition of a camera, computer, and other common computer peripherals. The Jetson computer can be adapted to control any CNC motor controller running GRBL firmware by sending commands over a USB serial connection. CNC kits have their own wiring schematics specific to make and model.

Table S1: Bill of Materials

Item	**Cost (CAD)**	**Type**
Jetson Orin Nano	$879	Computer
32 GB Fast SD Card	$50	Computer
1 TB NVMe SSD	$99	Computer
USB / Wireless Keyboard	$30	Computer
USB / Wireless Mouse	$30	Computer
USB-C Flash Drive or SSD	$40	Computer
DisplayPort to HDMI Adapter	$15	Computer
Monitor	$100	Computer
3 m HDMI Cable	$15	Computer
3 m USB-B to USB-A Cable	$15	Computer
Raspberry Pi HQ Camera	$78.28	Camera
Seeed Studio 10–100× C-Mount Lens	$101.50	Camera
Thermo Fisher Microscope Calibration Slide	$430	Camera
Generic Ring Light for Lens	$40	Camera
OpenBuilds BlackBox X32	$470	Frame
OpenBuilds Acro	$700	Frame
OpenBuilds NEMA17 Linear Actuator (250 mm)	$240	Frame
OpenBuilds Limit Switch	$15	Frame
Corner Brackets for Aluminum Extrusion	$30	Frame
20 mm × 20 mm × 250 mm V-Slot Aluminum Extrusion	$40	Frame
M3/M4/M5 Hammer Head Drop-In T-Nuts	$16	Frame
Metric Lock Nut Assortment	$14.99	Frame
Metric Nuts and Bolts Assortment	$29.99	Frame
Bullseye Level	$15	Tools
Total	**$3,494**	

Species	Type	Sample Type
Betula alleghaniensis	Angiosperm	Cookie, Core
Quercus rubra	Angiosperm	Cookie, Core
Alnus incana	Angiosperm	Cookie
Betula papyrifera	Angiosperm	Cookie
Betula populifolia	Angiosperm	Cookie
Abies amabilis	Gymnosperm	Core
Acer rubrum	Angiosperm	Core
Acer saccharum	Angiosperm	Core
Aesculus flava	Angiosperm	Core
Betula nigra	Angiosperm	Core
Carya glabra	Angiosperm	Core
Carya ovata	Angiosperm	Core
Callitropsis nootkatensis	Gymnosperm	Core
Populus deltoides	Angiosperm	Core
Pseudotsuga menziesii	Gymnosperm	Core
Quercus alba	Angiosperm	Core
Thuja plicata	Gymnosperm	Core
Tilia americana	Angiosperm	Core
Tsuga heterophylla	Gymnosperm	Core
Tsuga mertensiana	Gymnosperm	Core

Table S2: Tree species that have been scanned with TIM, their taxonomic group, and sample type.

Terms	Definition
X_0, Y_0, Z_0	The initial cartesian coordinates of the camera when the sample is saved by the user.
d_1	The relative distance from the camera to one edge of a leveled sample (Figure 3).
d_2	The relative distance from the camera to the opposite edge of a leveled sample (Figure 3).

Table S3: Mathematical notation used in this paper.

Filename	Species	Description
core-scan-ab-am-scalebar.tif	*Abies amabilis*	Scan of a 12 cm core 20,520.7 DPI.
core-scan-ac-ru-scalebar.tif	*Acer rubrum*	Scan of a 15 cm core 20,520.7 DPI.
core-scan-ca-ov-scalebar.tif	*Carya ovata*	Scan of a 25 cm core at 20,520.7 DPI.
core-scan-qu-ru-scalebar.tif	*Quercus rubra*	Scan of a 14 cm core at 20,520.7 DPI.
core-scan-ts-he-scalebar.tif	*Tsuga heterophylla*	Scan of a 18 cm core at 20,520.7 DPI.
cookie-scan-al-in-scalebar.tif	*Alnus incana*	Scan of a 40 mm x 35 mm cookie at 20,520.7 DPI.
cookie-scan-bet-all-scalebar.tif	*Betula alleghaniensis*	Scan of a 40 mm x 40 mm cookie at 20,520.7 DPI.
cookie-scan-bet-all-printed-ruler-scalebar.tif	*Betula alleghaniensis*	Scan of a 40 mm x 10 mm section of a cookie with with a paper ruler on top of the sample surface at 20,520.7 DPI.
cookie-scan-bet-all-printed-stage-micrometer-scalebar.tif	*Betula alleghaniensis*	Scan of a 40 mm x 10 mm section of a cookie with with a 1 mm stage micrometer on top of the sample surface at 20,520.7 DPI.
cookie-scan-bet-all-21259-scalebar.tif	*Betula alleghaniensis*	Partial scan of a cookie at 21,259.8 DPI.
cookie-scan-bet-pop-scalebar.tif	*Betula populifolia*	Scan of a 40 mm x 40 mm cookie at 20,520.7 DPI.
cookie-scan-q-a-scalebar.tif	*Quercus alba*	Scan of a 40 mm x 40 mm cookie at 20,520.7 DPI.
caliper-scalebar.tif	N/A	Scan of a caliper's millimeter graduations at 20,520.7 DPI.
stage-micrometer-20520.tif	N/A	Image of a 1 mm stage micrometer with 0.01 mm graduations used to find the scale of the scans at 20,520.7 DPI.
stage-micrometer-21259.tif	N/A	Image of a 1 mm stage micrometer with 0.01 mm graduations used to find the scale of the scan at 21,259.8 DPI.

Table S4: Filename of representative sample, species, and description of the file. The scans are stored in a Zenodo repository (https://doi.org/10.5281/zenodo.22215691).

Filename	Species	Control Points DPI	Control Points (mm)	Control Point Distance (Pixels)
core-scan-ac-sa-scalebar.tif	*Acer saccharum*	21,145	101.3	84,332.1
core-scan-be-ni-scalebar.tif	*Betula nigra*	21,108	100.8	83,769.9
core-scan-ps-me-scalebar.tif	*Pseudotsuga menziesii*	21,100	101.13	84,011.1
core-scan-qu-ru-scalebar.tif	*Quercus rubra*	21,169	101.03	84,200.1
core-scan-ts-me-scalebar.tif	*Tsuga mertensiana*	21,097	101.14	83,998.1
stage-micrometer-21080.tif	N/A	21,080	1.0	829.9

Table S5: Two control points were added to these scans at approximately the same distance with an ultra-fine tipped felt pen. The distance between the two points can be measured physically with digital calipers and digitally with an image annotation tool. The DPI measurements are similar to performing the same measurements on a stage micrometer at the same lens zoom. The scans are stored in a Zenodo repository (https://doi.org/10.5281/zenodo.23287846).

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